

Price-Segmentation and the U.S. Housing Boom

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Abstract

This paper presents novel stylized facts on U.S. house price growth by price segment in the 2000s housing boom. In each MSA studied, house price growth was generally higher for lower-priced, rather than higher-priced, houses during the boom. Likewise, in the bust, lower-priced houses experienced larger price declines. The similar pattern of house price growth by price segment is surprising given substantial variation in local fundamentals and boom-bust dynamics across MSAs. However, these results are consistent with existing findings that buyers of lower-priced houses are more sensitive to aggregate credit conditions.

Keywords: housing booms, house price growth, transaction data.

JEL: D14, D91, R21, R31

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1 Introduction

The U.S. housing boom of the 2000s varied in timing and magnitude across metropolitan statistical areas (MSAs) (Ferreira and Gyourko, 2023). Even within a single MSA, house price appreciation varied depending on whether a house was cheap or expensive (Landvoigt et al., 2015). To assess if this variation holds more broadly, this paper extends the analysis of Landvoigt et al. (2015) to additional MSAs and finds that house price growth tended to be higher for lower-priced, rather than higher-priced houses during the boom. In the bust, lower-priced houses lost their value faster than their higher-priced counterparts. Because these MSAs experienced different boom-bust dynamics—especially Cleveland, which had almost no boom—the broadly similar pattern of house price growth by price segment may seem surprising. While estimates vary in magnitude across MSAs—consistent with local differences—the estimates generally have the same sign, which is consistent with existing work on aggregate credit conditions. Specifically, Fidelman and Tapak (2026) and Nikolakoudis (2024) find that buyers of lower-priced houses are more sensitive to aggregate credit conditions, which are studied as an important driver of boom-bust dynamics by Favilukis et al. (2017), Kaplan et al. (2020), and Jacobson (2026).

To investigate how house price growth varies across MSAs and price segments, this paper estimates a statistical model of price changes developed by Landvoigt et al. (2015) using Zillow (2026) ZTRAX transaction data on repeat housing sales. Transaction-level data provides repeat sales prices of the same property, which, in turn, allows for the estimation of a common component of house price growth and a cross-sectional dispersion component across houses segmented by price. This model assumes a one-dimensional quality index so that the sale price of a house fully reflects its quality.

By expanding the analysis of Landvoigt et al. (2015) beyond San Diego, CA to include Phoenix, AZ and Cleveland, OH, this paper studies variation in house price growth in three MSAs that experienced different boom-bust patterns and characteristics of submarkets. San Diego is typically characterized by authors like Saiz (2010) as having a low elasticity of housing supply such that it is difficult—because of geography—to build more houses in response to higher house prices. On the other hand, Phoenix and Cleveland are both characterized as having geography that lends itself to a higher elasticity of housing supply. However, their fundamentals differ; Phoenix in the 2000s faced rapid housing and population growth, and Cleveland depopulation.¹ Despite these differences, house price growth tended to be higher for relatively lower-priced houses throughout the 2000s housing boom.

¹See Glaeser and Tobio (2007), Glaeser et al. (2008), and Glaeser and Gyourko (2005).

2 Estimates of House Price Growth

Transactions of repeat sales of single-family houses for the MSAs of Cleveland, OH; Phoenix, AZ; and San Diego, CA are obtained via Zillow (2026) ZTRAX assessment and transaction data accessed through the Bureau of Economic Analysis in 2019. Applying the cleaning steps detailed in Appendix A to arm's-length, non-foreclosed sales of residential properties made by owner-occupiers results in over 280,000 transactions of houses that are sold at least twice from 1999 to 2008.² Limiting the sample to repeat sales is essential because the statistical model estimates capital gains from the observed sales of the same property.

Capital gains of house i at time t in MSA j vary by their current log price $\log p_t^{i,j}$ where the idiosyncratic shocks $e_{t+1,t}^{i,j}$ have mean zero and are such that the law of large numbers holds in the cross section of houses. Because this paper assumes that there is a one-dimensional quality index, quality is reflected one-for-one in the house price at any given time t and there are no additional controls.³

The statistical model can be written as:

$$\log p_{t+1}^{i,j} - \log p_t^{i,j} = a_{t+1,t}^j + b_{t+1,t}^j \log p_t^{i,j} + e_{t+1,t}^{i,j} \quad (1)$$

Estimating the equation above would restrict the sample to houses that sold in successive years, such as a sale in 2000 and 2001, for example. To incorporate information from longer-dated sales, such as a house that sold in 2000 and again in 2003, for example, the above equation is iterated forward to capture capital gains between years $t+k$ and t for each $t = 1999, \dots, 2007$ and $k \in \{1, \dots, 2008 - t\}$.

$$\log p_{t+k}^{i,j} - \log p_t^{i,j} = a_{t+k,t}^j + b_{t+k,t}^j \log p_t^{i,j} + e_{t+k,t}^{i,j}$$

For $k \geq 2$, the above equation can be written in terms of $a_{t+1,t}^j$ and $b_{t+1,t}^j$.

$$\log p_{t+k}^{i,j} = a_{t+k,t+k-1}^j + \sum_{m=t+2}^{t+k} \prod_{\ell=m}^{t+k} (1 + b_{\ell,\ell-1}^j) a_{m-1,m-2}^j + \prod_{\ell=t+1}^{t+k} (1 + b_{\ell,\ell-1}^j) \log p_t^{i,j} \quad (2)$$

Equations (1) and (2) are estimated separately for each MSA $j \in \{\text{Cleveland, Phoenix, San Diego}\}$. Estimation is for $t = 1999, \dots, 2007$ via two-step GMM with a robust weighting matrix. The first-step residuals are used to estimate the weighting matrix for the second step, which ensures that the estimator is robust to heteroskedasticity. Estimates

²Appendix A also compares estimates of San Diego to those of Landvoigt et al. (2015).

³See Case and Shiller (1987) for the advantages of repeat sales approaches like the one used in this paper relative to hedonic approaches that control for property characteristics.

$\hat{a}_{t+k,t}^j$ and $\hat{b}_{t+k,t}^j$ can be obtained by adding/multiplying $\hat{a}_{t+1,t}^j$ and $\hat{b}_{t+1,t}^j$, respectively.

The slopes $\hat{b}_{t+1,t}^j$ are the coefficients of interest and test for cross-sectional dispersion in house price growth by price segment for each MSA j for any given pair of years $t + 1$ and t , conditional on the full sample of transactions between years $t + k$ and t . If $\hat{b}_{t+1,t}^j = 0$, then there is no cross-sectional dispersion and all houses in that MSA have the same capital gain regardless of their price. If $\hat{b}_{t+1,t}^j > 0$, then capital gains are relatively higher for more expensive houses in MSA j , which are those houses that have a higher sales price in year t . If $\hat{b}_{t+1,t}^j < 0$, relatively cheaper houses in MSA j have higher capital gains. Comparing the signs of $\hat{b}_{t+1,t}^j$ for the three MSAs representing different housing submarkets can provide insights on the role of local versus aggregate fundamentals. The intercepts $\hat{a}_{t+1,t}^j$ are the common component of capital gains for all housing transactions in MSA j .

Figure 1 shows the GMM estimates of the coefficients from equations (1) and (2) as the markers and 95% confidence intervals as the shaded bands for Cleveland, Phoenix, and San Diego. Estimates shown are those from one year t to the next year $t + 1$ such that $k = 1$. While these can be added/multiplied to obtain those that correspond to $k \geq 2$, those for $k = 1$ can be more readily compared to actual 12-month percentage change in house prices as shown in figure 2.⁴

The top panel 1a shows that throughout the housing boom, house price growth tended to be higher for lower-priced houses in all three MSAs, as shown by the estimates of the cross-sectional dispersion of capital gains across house price segments $\hat{b}_{t+1,t}^j$. In the subsequent bust, the estimates generally change sign when MSA-level house price indexes start to fall (as shown in figure 2) such that relatively lower-priced houses lost their value faster than their higher-priced counterparts. The sensitivity of relatively lower-priced houses to the boom-bust dynamics is consistent with other work (Ferreira and Gyourko, 2016; Mian and Sufi, 2009; Piazzesi and Schneider, 2016; Fidelman and Tapak, 2026; Nikolakoudis, 2024). These estimates show that a 1% less expensive house (relative to the average house) had at most a 0.1 percentage point larger capital gain in the boom ($t = 1999, \dots, 2005$) and a 0.2 percentage point larger capital loss in the bust ($t = 2006, 2007$).

The estimates of cross-sectional dispersion of capital gains $\hat{b}_{t+1,t}^j$ vary in magnitude across MSAs, which is consistent with local differences. Panel 1a points to lower-priced houses having relatively higher house price growth in San Diego than in Phoenix or Cleveland, on average. Because the housing supply elasticity is lower in San Diego than in Phoenix or Cleveland (according to Saiz, 2010), increased demand for less expensive houses was more

⁴The estimates for 2006 and 2007 are shown for completeness even though the housing bust was largely unanticipated. If these years were not included in estimation, then $\hat{b}_{t+1,t}^j$ and $\hat{a}_{t+1,t}^j$ could be even larger in magnitude than the values shown in figure 1.

likely to result in relatively higher price increases, and thus capital gains.⁵ Conversely, Phoenix’s near-zero dispersion suggests relatively uniform capital gains across house price segments. This can be attributed, in part, to a high share of speculative investors who—per Gao et al. (2020)—tended to be less sensitive to either local or aggregate fundamentals.⁶ Despite strikingly different house price dynamics compared to those in Phoenix and San Diego, Cleveland has estimates that are broadly similar, which is consistent with aggregate factors playing an important role in the housing boom.

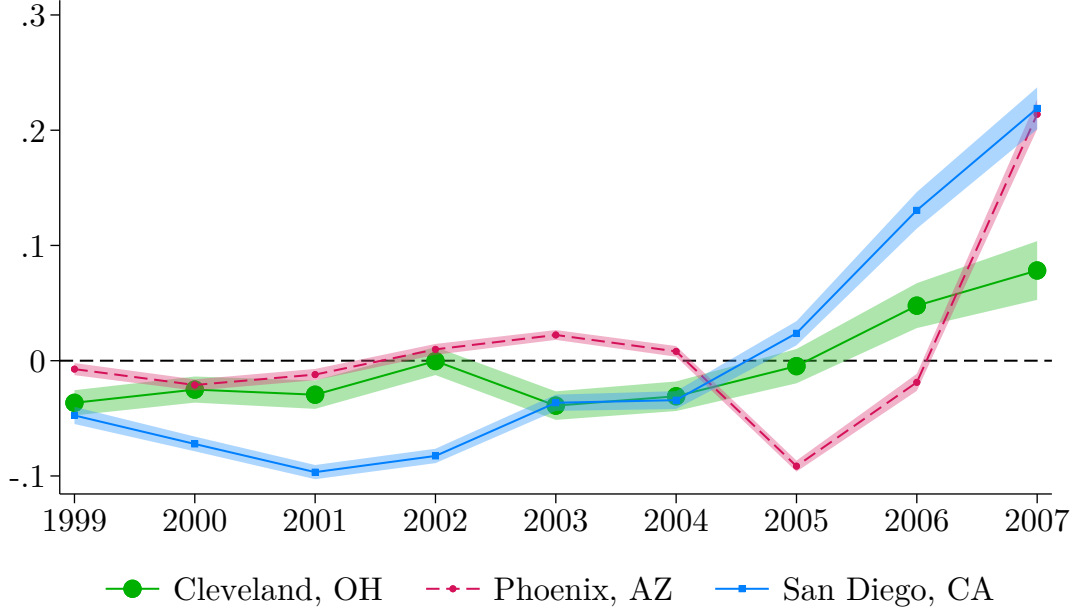
The estimated common component of house price growth, $\hat{a}_{t+1,t}^j$, shown in the bottom panel 1b, closely tracks the 12-month percentage change in house prices shown in figure 2. This alignment helps validate that the estimated data-generating process in equations (1) and (2) correctly reflects key attributes about regional house prices—specifically the early peak for San Diego, the later-arriving rapid surge for Phoenix, and virtually no boom for Cleveland. Notably, house price growth in Phoenix and Cleveland was similar and remained below that of San Diego until 2003, which likely reflects lower buyer incomes in these regions, as noted by Ferreira and Gyourko (2023). However, after 2003, capital gains in Phoenix and Cleveland diverge sharply, which is likely due to the influx of housing investors in the former, but not the latter.

3 Conclusion

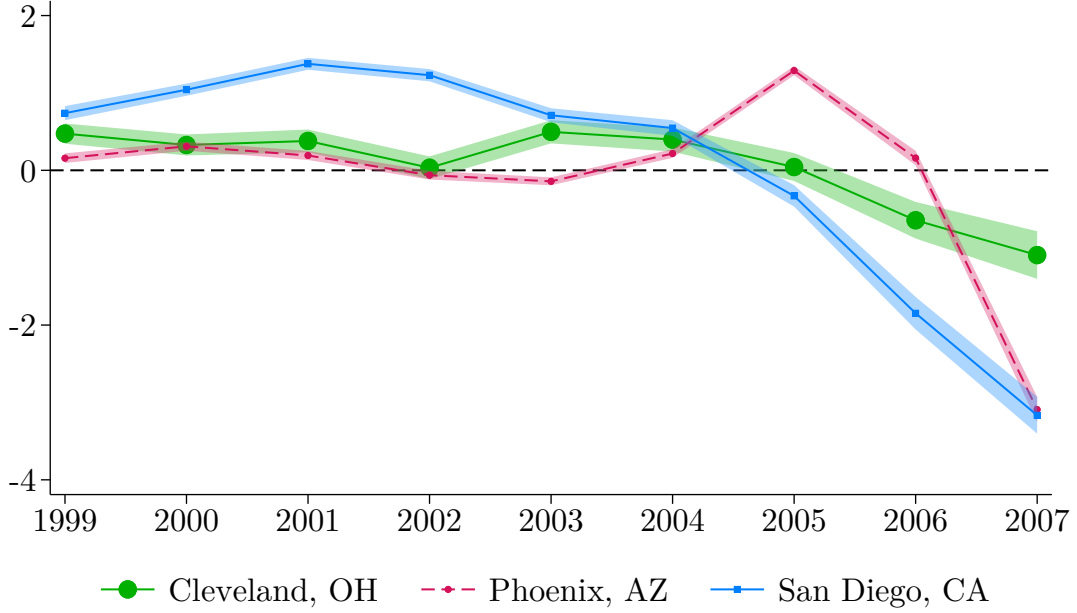
This paper documents novel stylized facts on U.S. house price growth by price segment in the 2000s housing boom to understand house price variation within and across MSAs. Estimating the statistical model of Landvoigt et al. (2015) on transaction-level housing data for the MSAs of Cleveland, OH; Phoenix, AZ; and San Diego, CA shows that lower-priced houses tended to have higher capital gains than their higher-priced counterparts. The estimates generally have similar signs even though the MSAs differ by local fundamentals and boom-bust dynamics. This finding is consistent with existing work that documents that buyers of lower-priced houses tend to be more sensitive to aggregate credit conditions (Fidelman and Tapak, 2026; Nikolakoudis, 2024). At the same time, differences in magnitudes of estimates are consistent with local variation.

⁵For discussions of the limitations of the estimates of Saiz (2010), see Aastveit et al. (2023), Oh et al. (2025), and Louie et al. (2025).

⁶Other studies on the role of investors include Graham (2024), Chinco and Mayer (2016), Mian and Sufi (2022), Bayer et al. (2021), Albanesi et al. (2022), Garriga et al. (2026a), and Garriga et al. (2026b). Excluding housing investors from the estimation sample could bias estimates for Phoenix and, to a lesser extent, San Diego if investors were concentrated in a particular price segment. However, focusing on owner-occupied properties facilitates comparisons with Cleveland where there were fewer investors.



(a) Cross-sectional Dispersion of House Price Growth ($\hat{b}_{t+1,t}^j$), Percentage Points



(b) Common Component of House Price Growth ($\hat{a}_{t+1,t}^j$), Percentage Points

Figure 1: Estimated cross-sectional dispersion and common component of house price growth with 95% confidence bands for repeat sales of single-family houses in Cleveland, OH; Phoenix, AZ; and San Diego, CA from the equations (1) and (2), $\log p_{t+1}^{i,j} - \log p_t^{i,j} = a_{t+1,t}^j + b_{t+1,t}^j \log p_t^{i,j} + e_{t+1,t}^{i,j}$ and $\log p_{t+k}^{i,j} = a_{t+k,t+k-1}^j + \sum_{m=t+2}^{t+k} \prod_{\ell=m}^{t+k} (1 + b_{\ell,\ell-1}^j) a_{m-1,m-2}^j + \prod_{\ell=t+1}^{t+k} (1 + b_{\ell,\ell-1}^j) \log p_t^{i,j}$, respectively, where $t = 1999, \dots, 2007$ and $k \in \{1, \dots, 2008 - t\}$. The data consist of 48,968 repeat sales in Cleveland, OH; 148,842 repeat sales in Phoenix, AZ; and 84,076 repeat sales in San Diego, CA via ZTRAX (Zillow, 2026).

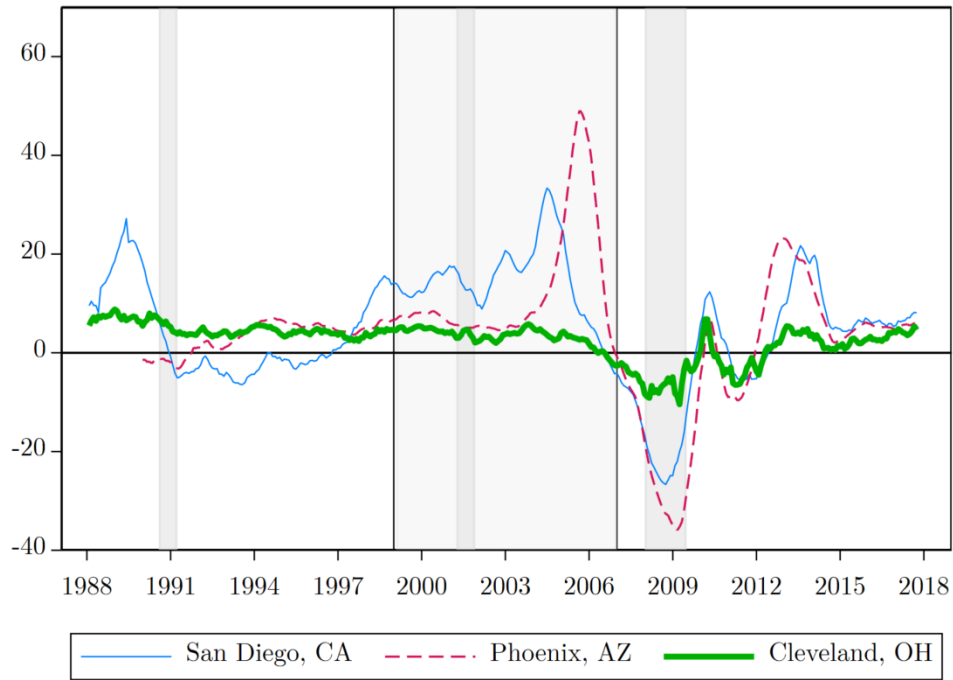


Figure 2: 12-month percentage change in house prices for select metropolitan statistical areas. Dark shaded bands are NBER recessions and the light shaded band between the two vertical lines is the U.S. housing boom period defined as 1999 to 2006 in this paper. Source: S&P Cotality Case-Shiller Home Price Indices, National Bureau of Economic Research.

A Appendix: Data

This section gives detailed cleaning steps of the Zillow (2026) ZTRAX transaction and assessment data to obtain a sample similar to that of Landvoigt et al. (2015) for the MSAs of Cleveland, OH; Phoenix, AZ; and San Diego, CA. The Zillow (2026) ZTRAX data is a panel of housing transactions and the cleaning steps can be grouped into those related to deeds, characteristics, and outliers.

First, to obtain housing market transactions, deeds (**documenttype**) that are not arm’s-length transfers of houses are dropped as are other non-arm’s-length deeds such as those indicating a partial sale of a house. Following Landvoigt et al. (2015), I keep grant deeds (**GRDE**), condo deeds (**CDDE**), corporate deeds (**CPDE**), and individual deeds (**IDDE**). Because the list of deeds denoting arm’s-length transactions is more exhaustive for Cleveland, OH and Phoenix, AZ, I delete entries for **documenttype** that are conservator’s deed (**CVDE**), deed in lieu of foreclosure (**DELU**), gift deed (**GFDE**), intrafamily transfer (**INTR**), partnership deed (**PTDE**), personal representative’s deed (**PRDE**), sheriff’s deed (**SHDE**), trustee’s deed (**TRFC**). Deeds that remain have values for **documenttype** that include administrator’s deed (**ADDE**), agreement of sale (**AGSL**), bargain and sale deed (**BSDE**), condominium deed (**CDDE**), court order/action (**COCA**), corporation deed (**CPDE**), correction deed (**CRDE**), deed (**DEED**), fiduciary deed (**FDDE**), guardian’s deed (**GDDE**), grant deed (**GRDE**), individual deed (**IDDE**), joint tenancy deed (**JTDE**), land contract (**LDCT**), other (**OTHR**), quitclaim deed (**QCDE**), re-recorded deed (**RRDE**), tax deed (**TXDE**), and warranty deed (**WRDE**).

Second, deeds are dropped based on buyer or house characteristics. Deeds without a latitude or longitude are dropped as are deeds that transfer multiple parcels as identified by the APN. Second homes and trailers are dropped as are foreclosed properties. Buyers that are not a couple or single person are dropped eliminating buyers that are a corporation or partnership (**C0,PT**), a trust (**FT,IT,LV,RL,RT,TE**), or a beneficiary (**BF**).⁷

Single-family houses are denoted by the **propertylanduse** variable from the assessment data⁸ and observations that are kept include single-family residences (**RR101**), condominiums (**RR106**)⁹, cooperatives (**RR107**), row houses (**RR108**), planned unit developments (**RR109**),

⁷This is achieved by keeping observations with **buyercode** equal to domestic partners (**DP**), formerly known as (**FK**), her husband (**HH**), husband and wife (**HW**), individual (**ID**), married man (**MM**), minor (**MN**), married person (**MP**), married woman (**MW**), single man (**SM**), single person (**SP**), single woman (**SW**), unmarried man (**UM**), unmarried woman (**UW**), widowed (**WW**) and dropping those with **buyercode** equal to affiant (**AF**), borrower or trustor in default (**BR**), estate (**ES**), executor (**EX**), government (borough, city, village, etc.) (**GV**), surviving joint tenant (**SJ**), personal representative (**PR**), agent (**AG**), not provided (**NP**).

⁸The **propertylanduse** variable from the assessment data and the **propertyusestndcode** variable from the transaction data are mostly but not always consistent. I use the assessment data as the main variable to denote property use because it has fewer blank observations and is more stable across time.

⁹Although it may be debatable whether or not condominiums should be included as single-family houses,

bungalows (RR113), zero lot lines (RR114), manufactured, modular, and prefabricated houses (RR115), patio homes (RR116), garden homes (RR119), condominiums (RR120), and inferred single-family houses (RR999). Dropped observations have a `propertylanduse` variable from the assessment data equal to rural residences including farms/productive land (RR102), mobile homes (RR103), residential common areas (RR110), time shares (RR111), seasonal, cabin, vacation residences (RR112), residential parking garages (RR117), and other improvements (RR118). Observations from the transaction data are also dropped. These include observations where the `propertyusestndcode` variable equals agricultural (AG), apartment (AP), commercial (CM), mobile homes (MB), mixed use (MX), unimproved (UL), multifamily (MF).

Lastly, to control for outliers, transactions with prices below \$15,000, combined loan-to-value ratios (first plus second mortgage) above 120 %, and annualized capital gains above 50% are dropped. To avoid the influence of house flipping, all pairs of sales that are less than 180 days apart are dropped. Some properties that were sold twice in the same year, but more than 180 days apart remain in the sample. If the property was sold more than twice in the same year, then these transactions are dropped. If the property was sold in another year then the earlier sale date in the year is dropped. Given that house prices are rising throughout this period, keeping the later sale should bias capital gains downward. Given the high house price growth observed at the MSA-level in Phoenix, AZ, robustness checks were run to increase the annualized capital gain threshold from 50% to 60% and 70% with little change to the estimates. Similarly, second homes were included in a separate robustness check with little alteration to the estimates.

Table 1 compares the estimates of common and cross-sectional capital gains for San Diego, CA to those of Landvoigt et al. (2015). Overall, the estimated coefficients from equations (1) and (2) resemble those of Landvoigt et al. (2015). A few minor discrepancies likely arise from differences in the source data used (Trulia vs. Zillow). My sample is larger than theirs with 84,076 repeat sales compared to their 70,315.

dropping condominiums decreases the number of observations from 84,076 to 53,463 for San Diego, CA and the estimates of capital gains differ to a larger extent from those of Landvoigt et al. (2015) shown in table 1.

	2000	2001	2002	2003	2004	2005	2006	2007
$a_{t+1,t}^{LPS}$	1.29 (0.04)	1.41 (0.04)	1.30 (0.04)	0.87 (0.05)	0.60 (0.06)	-0.56 (0.07)	-1.09 (0.10)	-3.18 (0.12)
$b_{t+1,t}^{LPS}$	-0.09 (0.003)	-0.10 (0.003)	-0.09 (0.003)	-0.05 (0.004)	-0.04 (0.004)	0.04 (0.010)	0.07 (0.010)	0.22 (0.010)
$a_{t+1,t}^{author}$	1.04 (0.04)	1.37 (0.04)	1.23 (0.04)	0.71 (0.05)	0.55 (0.05)	-0.33 (0.07)	-1.84 (0.10)	-3.16 (0.12)
$b_{t+1,t}^{author}$	-0.07 (0.003)	-0.10 (0.003)	-0.08 (0.003)	-0.04 (0.004)	-0.03 (0.004)	0.02 (0.005)	0.13 (0.008)	0.21 (0.009)

Table 1: Upper panel contains estimates from Landvoigt et al. (2015) (LPS) of 70,315 repeat sales in San Diego County during the years 1999-2008 using transaction data from Trulia. The lower panel contains a replication of their estimates using ZTRAX (Zillow, 2026) assessment and transaction data for 84,076 repeat sales in San Diego County. The numbers in parentheses are standard errors.

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